Experiment 7a:

CODE:

```
 GNU nano 8.7.1
#include <stdio.h>
int main() {
    long long counter = 0;
    while (1) {
        counter++;
        if (counter % 100000000 == 0) {
            printf("CPU task running: %lld\n", counter);
        }
    }
    return 0;
}
```

OUTPUT:

```
tofiq@Tofiq:~$ pgrep -a exp7a
5152 ./exp7a
tofiq@Tofiq:~$ top -p 5152
```

```
top - 06:35:24 up 15 min,  1 user,  load average: 0.57, 0.41, 0.31
Tasks:   1 total,   1 running,   0 sleeping,   0 stopped,   0 zombie
%Cpu(s): 12.4 us,  2.1 sy,  0.0 ni, 83.3 id,  0.8 wa,  0.0 hi,  1.4 si,  0.0 st
MiB Mem :   4371.1 total,   1214.5 free,   1253.2 used,   2181.8 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   3117.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   5152 tofiq     20   0    2768   1748   1648 R  98.0   0.0   0:54.73 exp7a
```

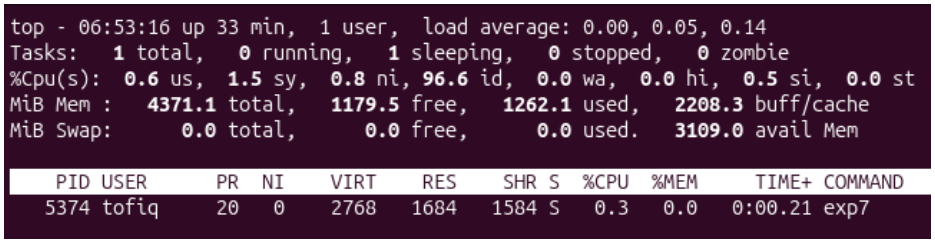
Experiment 7b:

CODE:

```
 GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main() {
    FILE *fp;
    printf("Starting I/O task...\n");
    while (1) {
        fp = fopen("dummy.txt", "w");
        if (fp != NULL) {
            fprintf(fp, "Writing disk data\n");
            fclose(fp);
            printf("File operation done\n");
        }
        sleep();
    }
    return 0;
}
```

OUTPUT:

```
tofiq@Tofiq:~$ pgrep -a exp7
5374 ./exp7
tofiq@Tofiq:~$ top -p 5374
```


top - 06:53:16 up 33 min, 1 user, load average: 0.00, 0.05, 0.14
Tasks: 1 total, 0 running, 1 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.6 us, 1.5 sy, 0.8 ni, 96.6 id, 0.0 wa, 0.0 hi, 0.5 si, 0.0 st
MiB Mem : 4371.1 total, 1179.5 free, 1262.1 used, 2208.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3109.0 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
5374 tofiq 20 0 2768 1684 1584 S 0.3 0.0 0:00.21 exp7